# Chapter 2:
# Set Theory

## 2.1 Set Theory Preliminaries

We begin with our common understanding of the meaning of the word set.

**Definition.** *A <u>set</u> is a collection of unique objects. Objects contained in a set are called <u>elements</u>. When $A$ is a set and $x$ is an element of $A$ we write $x \in A$.*

**Example.**

The elements of a set are unordered.

**Example.**

**Example.** Let $S$ be the set of all things that are not elephants.

**Russell's Paradox:** *Let $P$ be the set of all sets that are not elements of themselves.*

**Question.** Is $P \in P$?

**Barber's Paradox:** *There is a barber who shaves all those, and only those, who do not shave themselves.*

**Question.** Does the barber shave himself?

**Definition. (Vector space axioms)**

*A* <u>*vector space*</u> *is a set* $V$ *together with addition and scalar multiplication so that the following properties hold true:*

1. *For all* $x, y \in V$, $x + y \in V$.
2. *For all* $x \in V$ *and* $a \in \mathbb{R}$, $ax \in V$.
3. $x + y = y + x$ *for all* $x, y \in V$.
4. $(x + y) + z = x + (y + z)$ *for all* $x, y, z \in V$.
5. *There exists an element* $0 \in V$ *so that* $x + 0 = x$ *for all* $x \in V$.
6. *For each* $x \in V$ *there exists* $x' \in V$ *so that* $x + x' = 0$.
7. $1x = x$ *for all* $x \in V$.
8. $a(x + y) = ax + ay$ *for all* $x, y \in V$ *and* $a \in \mathbb{R}$.

*We call the elements of a vector space* <u>*vectors*</u>.

**Definition 2.1. (Set Axioms)**

- **Equality:** *Two sets are equal when they have the same elements.*
- **Specification:** *If $U$ is a set and $p(x)$ is a condition on $U$ expressed in first-order logic, then the collection of all elements of $U$ for which $p(x)$ is true is a set.*
- **Pairing:** *If $A$ and $B$ are sets, then there is a set that contains $A$ and $B$ as elements.*
- **Union:** *For any collection of sets $C$, there exists a set that contains all elements that belong to at least one set in $C$.*
- **Power Set:** *If $A$ is a set then there is a set that contains all of the subsets of $A$.*
- **Replacement:** *The range of a function is a set.*
- **Inductive Set:** *There exists a set $A$ so that every $x \in A$ is a set, $\emptyset \in A$, and if $x \in A$ then $x \cup \{x\} \in A$.*
- **Foundation:** *Every non-empty set $A$ contains an element $y$ so that $A$ and $y$ have no common elements.*
- **Axiom of Choice:** *Given any non-empty collection of sets, there is a set that has exactly one element from each set.*

Let's look at these axioms more carefully.

**Axiom of Equality**

**Definition 2.2.** *Let $A$ and $B$ be sets. We say $A$ and $B$ are* <u>*equal*</u> *when they have the same elements. In other words, we say $A$ and $B$ are equal when the following statement is true:*

$$(\forall x)\ x \in A \Leftrightarrow x \in B$$

*When $A$ and $B$ are equal, we write $A = B$.*

**Definition 2.3.** *Let $A$ and $B$ be sets. We say $B$ is a* <u>*subset*</u> *of $A$ when every element of $B$ is also an element of $A$. In other words, we say $B$ is a subset of $A$ when the following statement is true:*

$$(\forall x)\ x \in B \Rightarrow x \in A$$

*When $B$ is a subset of $A$, we write $B \subseteq A$.*

*When $B$ is a subset of $A$, and $A \neq B$, we say $B$ is a proper subset of $A$ and we write $B \subsetneq A$.*

**Aside.** *Some authors use the notation $\subset$ in place of $\subsetneq$. Feel welcome to use either in your submitted work.*

## Axiom of Union

**Definition 2.4.** *Let $A$ and $B$ be sets. The* <u>*union*</u> *of $A$ and $B$ is the set containing all elements of $A$ and all elements of $B$. We denote the union of $A$ and $B$ as $A \cup B$. That is*

$$A \cup B = \{x \mid (x \in A) \vee (x \in B)\}$$

## Axiom of Specification

The Axiom of Specification tells us that we may use conditions to specify membership in a set. That is, if $U$ is a set and $p(x)$ is a condition on $U$, then the collection of all elements of $U$ for which $p(x)$ is true forms a set. We denote such a set as

$$\{x \in U \mid p(x)\}$$

This method of specifying a set is usually referred to as <u>*set-builder notation*</u>.

**Definition 2.5.** *Let $U$ be a set and let $A$ and $B$ be subsets of $U$. The* <u>*complement*</u> *of $B$ in $A$ is the set containing all elements $A$ which are not element of $B$. We denote the complement of $B$ in $A$ as $A \setminus B$. That is*

$$A \setminus B = \{x \in A \mid x \notin B\}$$

**Definition 2.6.** *Let $A$ be a set. When $A$ contains no elements, we say $A$ is the* <u>*empty set*</u>*. When $A$ is the empty set, we write $A = \{\,\}$.*

*Aside. Some authors use the notation $\emptyset$ to denote the empty set. Feel welcome to use $\emptyset$ in your submitted work.*

**Definition 2.7.** *Let $U$ be a set and let $A$ and $B$ be subsets of $U$. The* <u>*intersection*</u> *of $A$ and $B$ is the subset of $U$ that contains all elements common to both $A$ and $B$. We denote the intersection of $A$ and $B$ as $A \cap B$. That is*

$$A \cap B = \{x \in U \mid (x \in A) \wedge (x \in B)\}$$

**Definition.** *Let $A$ and $B$ be sets. For each $a \in A$ and $b \in B$ we define the notation $(a, b)$ to be the set:*

$$(a, b) = \{a, \{a, b\}\}$$

*We call $(a, b)$ an <u>ordered pair</u>.*

We often regard $(a, b)$ as having 1st coordinate $a$ and 2nd coordinate $b$.

Notice that the set $(a, b)$ contains two elements, $a$ and $\{a, b\}$, and that $a \in \{a, b\}$.

Using this, we can show that $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.

**Definition 2.10.** *Let $A$ and $B$ be sets. The* <u>*Cartesian product*</u> *of $A$ and $B$ is the set of all ordered pairs whose first entry is in $A$ and whose second entry is in $B$. We denote the Cartesian product of $A$ and $B$ as $A \times B$. That is*

$$A \times B = \{(a, b) \mid (a \in A) \wedge (b \in B)\}$$

**Theorem 2.8.** *Let $A$ and $B$ be sets. We have $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.*

Recall that given statements $p$ and $q$

$$p \Leftrightarrow q \quad \equiv \quad (p \Rightarrow q) \wedge (q \Rightarrow p)$$

**Theorem.** *Let $A$, $B$, and $C$ be sets. Then* $(B \setminus A) \cap C = (B \cap C) \setminus A$.

Notice that we need to show two things:

*Proof*. Let $A$, $B$, and $C$ be sets. Let's show that $(B \setminus A) \cap C \subseteq (B \cap C) \setminus A$.

1.

2.

3.

4.

5.

6.

7.

8.

Exercise. Show that $(B \cap C) \setminus A \subseteq (B \setminus A) \cap C$.

## 4.1 Functions and Relations

We can use the Cartesian product to formalize the notion of a *relation* between sets.

**Example.** The University of Melbourne has a database of the students enrolled in various subjects.

Let $S$ be the set of all subjects at the University and let $P$ be the set of all students. Then the University's database can be represented as

**Definition 4.1.** *Let $A$, $B$, and $R$ be sets. We say that $R$ is a* <u>*relation*</u> *between $A$ and $B$ when $R$ is a subset of $A \times B$.*

We can regard functions as a type of relation which assigns to each input an output.

**Example.** Consider the set

$$A = \{(z, z^2) \mid z \in \mathbb{R}\}$$

**Definition 4.2.** *Let $A$ and $B$ be sets and $f$ be a relation on $A$ and $B$.*

*We say $f$ is a <u>function</u> from $A$ to $B$ when each element $x \in A$ appears as the first entry of an ordered pair $(x, y)$ for exactly one element $(x, y) \in f$.*

*When $f$ is a function from $A$ to $B$, we write $f: A \to B$.*

*We call $A$ the <u>domain</u> of $f$ and $B$ the <u>codomain</u> of $f$.*

*We say that $R$ is a relation between $A$ and $B$ when $R$ is a subset of $A \times B$.*

**Definition 4.3.** *For each $(x, y) \in f$ we call*

- *$x$ a <u>preimage</u> of $y$ and*
- *$y$ the <u>image</u> of $x$.*

*We write $f(x) = y$.*

**Definition 4.4.** *Let $f: A \to B$ be a function and $y \in B$.*

*The <u>pre-image</u> of $y$ is the set $f^{-1}(b) = \{x \in A \mid f(x) = y\}$.*

**Definition 4.5.** *The <u>range</u> of $f$ is the set* $\mathrm{range}(f) = \{f(x) \in B \mid x \in A\}$.

**Example.** Let $f: \mathbb{R} \to \mathbb{R}$ be the function defined by $f(z) = z^2$ for all $z \in \mathbb{R}$.

## 2.2 Ordered and Bounded Sets

Take a moment to draw a picture of $\mathbb{Z}$, the set of all integers.

**Definition 2.11.** *Let $S$ be a set. An* <u>*order*</u> *on $S$ is a relation, denoted $<$, so that the following two statements are true:*

(O1) *for all $x, y \in S$ exactly one of the following is true:*

- $x < y$
- $y < x$
- $x = y$

(O2) *for all $x, y, z \in S$, if $x < y$ and $y < z$, then $x < z$.*

*Additionally, we let $x > y$ denote the statement $y < x$. We let $x \leq y$ and $y \geq x$ both denote the statement $(x < y) \vee (x = y)$.*

**Example.** $S = \mathbb{R}$ with the usual inequality $<$ is an ordered set.

**Example.** A set $S = \{a\}$ with exactly one element is ordered.
As a subset of $S \times S$, the order $<$ is the empty set.

**Example.** Let $S = \{1,2,3\}$. Which of the following relations $<$ is an order on $S$?

- $1 < 2, \quad 1 < 3, \quad 2 < 3$

- $2 < 1, \quad 3 < 1, \quad 2 < 3$

- $1 < 2, \quad 2 < 3, \quad 3 < 1$

**Examples.**

The function $f(x) = \frac{1}{x}$ is bounded below by 0 on the domain $x > 0$.

The function $f(x) = \frac{1}{x}$ is not bounded above on the domain $x > 0$.

**Definition 2.12.** *Let $S$ be an ordered set and let $A$ be a subset of $S$. We say that $A$ is* <u>*bounded below*</u> *in $S$ when there exists $\beta \in S$ so that the following statement is true*

$$(\forall x \in A)\ \ \beta \leq x$$

*We say that $\beta$ is a* <u>*lower bound*</u> *for $A$ in $S$.*

**Definition 2.13.** *Let $S$ be an ordered set and let $A$ be a subset of $S$. We say that $A$ is* <u>*bounded above*</u> *in $S$ when there exists $\beta \in S$ so that the following statement is true*

$$(\forall x \in A)\ \ x \leq \beta$$

*We say that $\beta$ is an* <u>*upper bound*</u> *for $A$ in $S$.*

**Example.** $A = \left\{ \frac{1}{r} \middle| r \in \mathbb{R} \text{ and } r > 0 \right\}$ in $S = \mathbb{R}$

**Definition 2.14.** *Let $S$ be an ordered set and let $A \subseteq S$.*

*Let $U_A$ be the set of all upper bounds of $A$ in $S$.*

*We say that $\alpha \in U_A$ is a <u>least upper bound</u> of $A$ in $S$ when*

$$(\forall \beta \in U_A) \quad \alpha \leq \beta$$

*We also call the least upper bound $\alpha$ the <u>supremum</u> of $A$ in $S$ and write*

$$\sup A = \alpha$$

**Definition 2.15.** *Now let $L_A$ be the set of all lower bounds of $A$ in $S$.*

*We say that $\alpha \in L_A$ is a <u>greatest lower bound</u> of $A$ in $S$ when*

$$(\forall \beta \in L_A) \quad \alpha \geq \beta$$

*We also call the greatest lower bound $\alpha$ the <u>infimum</u> of $A$ in $S$ and write*

$$\inf A = \alpha$$

**Example 2.16.** $A = \{1,2,3,4,10\}$ in $S = \mathbb{N}$

**Example 2.17.** $A = \{x \in \mathbb{R} \mid x^2 \leq 2\} \cup \{x \in \mathbb{R} \mid x < 0\}$ in $S = \mathbb{R}$

**Example 2.18.** $A = \{x \in \mathbb{Q} \mid x^2 \leq 2\} \cup \{x \in \mathbb{Q} \mid x < 0\}$ in $S = \mathbb{Q}$

Example suggests:

*There is at least one rational number*
*between every rational and every irrational number.*

Let $S$ be an ordered set and let $A \subseteq S$. Let $\alpha \in S$.

Let $U_A$ be the set of all upper bounds of $A$ in $S$.

To prove $\alpha = \sup A$, we must prove two things:

1. $(\forall x \in A)\ \ x \leq \alpha$ ($\alpha$ is an upper bound for $A$)
2. $(\forall \beta \in U_A)\ \ \alpha \leq \beta$ ($\alpha$ is not bigger than any other upper bound for $A$)

Let $L_A$ be the set of all lower bounds of $A$ in $S$.

To prove $\alpha = \inf A$, we must prove two things:

1. $(\forall x \in A)\ \ \alpha \leq x$ ($\alpha$ is an lower bound for $A$)
2. $(\forall \beta \in L_A)\ \ \beta \leq \alpha$ ($\alpha$ is not bigger than any other lower bound for $A$)

**Example.** $A = \{x \in \mathbb{R} \mid 1 \leq x < 2\}$

**Claim.** $\inf A = 1$

First we show that $(\forall x \in A)\ \ x \leq 1$.

1.

2.

Next we show that $(\forall \beta \in L_A)\ \ \beta \leq 1$.

1.

2.

3.

**Example.** $A = \{x \in \mathbb{R} \mid 1 \leq x < 2\}$

**Claim.** $\sup A = 2$

First we show that $(\forall x \in A)\ \ x \leq 2$.

1.

2.

3.

Next we show that $(\forall \beta \in U_A)\ \ 2 \leq \beta$.

1.

2.

**Example.** $A = \{x \in \mathbb{R} \mid 1 \leq x < 2\}$

**Claim.** $\sup A = 2$

Let's try again… Next we show that $(\forall \beta \in U_A)\ \ 2 \leq \beta$. We proceed by contraction.

1.
2.
3.
4.
5.
6.
7.
8.
9.
10.

**Theorem 2.19.** *Let $A \subseteq \mathbb{R}$ be bounded above and non-empty.*

*Let $\gamma \in \mathbb{R}$ be an upper bound of $A$ in $\mathbb{R}$.*

$\sup A = \gamma$ *if and only if for every $\epsilon > 0$ there exists $x \in A$ such that $x > \gamma - \epsilon$.*

**Completeness Axiom.** *Every subset $A \subseteq \mathbb{R}$ that is bounded above in $\mathbb{R}$ has a least upper bound in $\mathbb{R}$.*

# 3.3 Sets and Operations

**Definition 2.20.** *Let $F$ be a set equipped with two well-defined binary operations, addition* $(+)$ *and multiplication* $(\times)$. *We say that $F$ is a* <u>*field*</u> *when the following statements are true:*

Moreover, the following statements are true:

| | | |
|---|---|---|
| (A1) | $(\forall x, y \in F)\ \ x + y \in F$ | (*addition is closed*) |
| (A2) | $(\forall x, y \in F)\ \ x + y = y + x$ | (*addition is commutative*) |
| (A3) | $(\forall x, y, z \in F)\ \ (x + y) + z = x + (y + z)$ | (*addition is associative*) |
| (A4) | $(\exists 0 \in F)\ \ [(\forall x \in F)\ \ x + 0 = x]$ | (*additive identity*) |
| (A5) | $(\forall x \in F)\ \ [(\exists y \in F)\ \ x + y = 0]$ | (*additive inverse*) |
| (M1) | $(\forall x, y \in F)\ \ xy \in F$ | (*multiplication is closed*) |
| (M2) | $(\forall x, y \in F)\ \ xy = yx$ | (*mult. is commutative*) |
| (M3) | $(\forall x, y, z \in F)\ \ (xy)z = x(yz)$ | (*mult. is associative*) |
| (M4) | $(\exists 1 \in F \setminus \{0\})\ \ [(\forall x \in F)\ \ 1x = x]$ | (*multiplicative identity*) |
| (M5) | $(\forall x \in F \setminus \{0\})\ \ [(\exists y \in F \setminus \{0\})\ \ xy = 1]$ | (*multiplicative inverse*) |
| (D) | $(\forall x, y, z \in F)\ \ x(y + z) = xy + xz$ | (*distributive property*) |

**Axioms of equality.** In logic, the equals sign follows the following two rules:

- *Law of identity:* $(\forall a)\ \ a = a$

- *Substitution:* Let $\phi(x)$ be any logical condition.
  If $a = b$ and $\phi(a)$ is true, then $\phi(b)$ is true.

  Consequently, the following is true. Let $f: A \to B$ be any function.
  If $a, b \in A$ such that $a = b$, then $f(a) = f(b)$.

**Binary operators.** The assumption that addition (and multiplication) is a <u>well-defined binary operation</u> means that addition is a function

$$\begin{aligned} &\text{Sum} : F \times F \to F \\ &\text{Sum}(x, y) = x + y \qquad \text{for each } x, y \in F. \end{aligned}$$

By substitution, for each $x, y, z \in F$

$$\begin{aligned} x = y \quad &\Longrightarrow \quad \text{Sum}(x, z) = \text{Sum}(y, z) \\ &\Longrightarrow \qquad\quad x + z = y + z \end{aligned}$$

**Note.** The assumption that addition (and multiplication) is a <u>well-defined binary operation</u> means that addition is a function

$$\text{Sum} : F \times F \to F$$

$$\text{Sum}(x, y) = x + y \qquad \text{for each } x, y \in F.$$

In particular, for each $x, y, z \in F$

$$\begin{aligned} x = y \quad &\Longrightarrow \quad \text{Sum}(x, z) = \text{Sum}(y, z) \\ &\Longrightarrow \qquad x + z = y + z \end{aligned}$$

**Definition 2.21.** *Let $F$ be a field.*

*We call* $0$ *the* <u>*additive identity*</u> *of* $F$.

*Given* $x, y \in F$ *such that* $x + y = 0$, *we call* $y$ *the* <u>*additive inverse*</u> *of* $x$. *When* $y$ *is the additive inverse of* $x$, *we denote* $y$ *as* $-x$.

**Definition 2.22.** *We call* $1$ *the* <u>*multiplicative identity*</u> *of* $F$.

*Given* $x, y \in F$ *such that* $xy = 1$, *we call* $y$ *the* <u>*multiplicative inverse*</u> *of* $x$. *When* $y$ *is the* <u>*multiplicative*</u> *inverse of* $x$, *we denote* $y$ *as* $1/x$ *or* $x^{-1}$.

**Example.** $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{F}_2 = \{0,1\}$ are well-known fields.

**Example.** $\mathbb{Z}$ has addition and multiplication operators, but $\mathbb{Z}$ is *not* a field since it does not contain the multiplicative inverses of every non-zero integer (as in (M5)).

**Example.** Let

$$M = \left\{ \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \middle| \, a, b \in \mathbb{R} \right\}$$

Our experience in Linear Algebra tells us how to add and multiply elements of $M$.

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} + \begin{bmatrix} c & -d \\ d & c \end{bmatrix} = \begin{bmatrix} a+c & -(b+d) \\ b+d & a+c \end{bmatrix}$$

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} c & -d \\ d & c \end{bmatrix} = \begin{bmatrix} ac-bd & -(ad+bc) \\ ad+bc & ac-bd \end{bmatrix}$$

One can verify that $M$ is a field.

Clearly $M$ is closed under addition and multiplication.

The additive identity 0 in $M$ is the $2 \times 2$ zero matrix $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

The multiplicative identity 1 in $M$ is the $2 \times 2$ identity matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

**Theorem 2.23.** *Let $F$ be a field and let $x, y, z \in F$. If $x + y = x + z$, then $y = z$.*

*Idea of proof.*

*Proof.*

**Theorem 2.24.** *Let $F$ be a field and let $x, y, z \in F$ with $x \neq 0$. If $xy = xz$, then $y = z$.*

Consider the sequence of subsets

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

The natural numbers $\mathbb{N}$ is informally defined by

One can formally define $\mathbb{N}$ using the Peano Axioms to specify the properties of the successor function $S(n) = n + 1$.

Using Set Theory, one can represent $\mathbb{N}$ by defining $S(A) = A \cup \{A\}$ for each set $A$ and writing

Consider the sequence of subsets

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

The natural numbers $\mathbb{N}$ is not a field because it does not contain additive or multiplicative inverses.

The integers $\mathbb{Z}$ is defined as the set containing every natural number $n$ and its additive inverse $-n$.

$\mathbb{Z}$ is still not a field because it does not contain multiplicative inverses.

The rational numbers $\mathbb{Q}$ is defined as the set of all ratios $a/b$ where $a, b \in \mathbb{Z}$ and $b \neq 0$. We require that

$$\frac{a}{b} = \frac{c}{d} \quad \text{if and only if} \quad ad = bc.$$

$\mathbb{Q}$ is a field.

**Theorem.** *The real numbers* $\mathbb{R}$

- *are an ordered field,*
- *contain* $\mathbb{Q}$ *as a subfield, and*
- *have the property that every non-empty subset* $A \subseteq \mathbb{R}$ *that is not bounded above in* $\mathbb{R}$ *has a supremum in* $\mathbb{R}$.